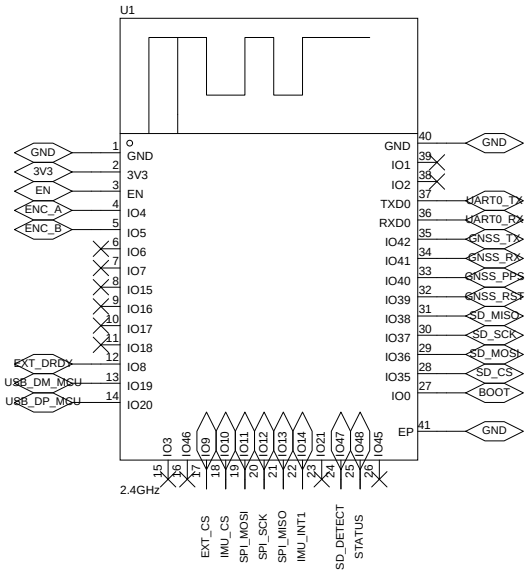
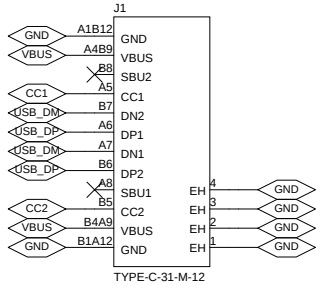
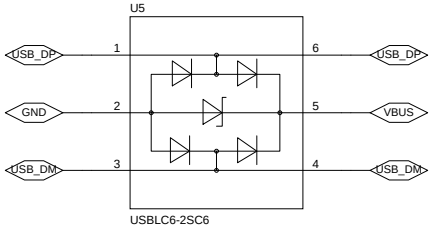
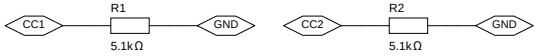
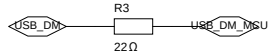
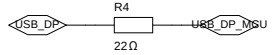


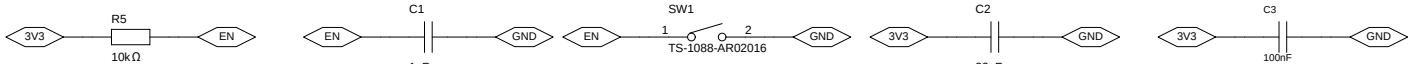


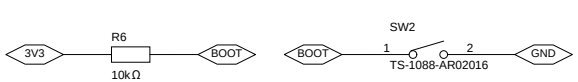














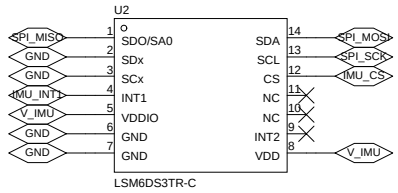
01 MCU / USB / BOOT

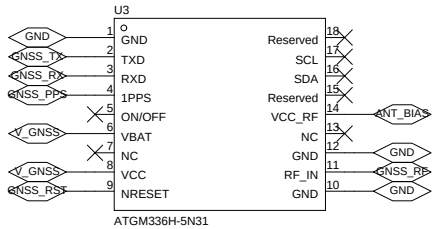
ESP32-S3-WROOM-1-N8 (no PSRAM); USB native FS

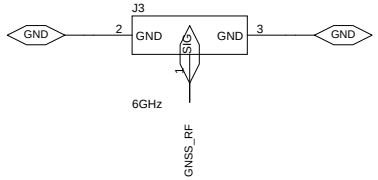
BOOT: hold SW2, tap SW1 RESET, release SW2

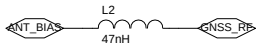
PCB bottom TP1=TX0 TP2=RX0 TP3=EN TP4=BOOT

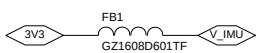
原理图	Schematic1	创建日期	2026-10-16
		更新日期	2026-10-16
板子	Board1	图页	01_MCU_USB
绘制		INSmini_S3_R1	
审阅			
		版本	尺寸	页 1 共 4
嘉立创EDA		V1.0	A4	嘉立创EDA

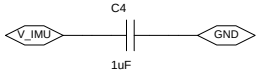






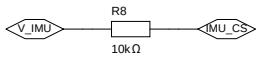


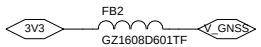


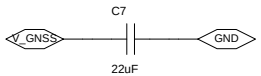


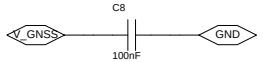












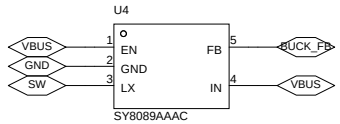
02 IMU / GNSS

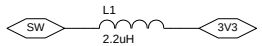
LSM6DS3TR-C SPI + INT1 / FIFO; ATGM336H-5N31 UART + PPS

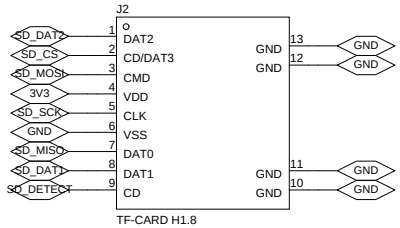
J3: external 3.3 V active GNSS antenna; bias max 50 mA

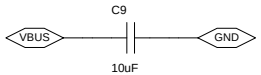
V_GNSS: filtered 3V3; VBAT tied to supply, cold start after power-off

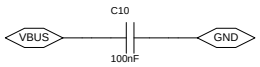
原理图	Schematic1	创建日期	2026-10-16
		更新日期	2026-10-16
板子	Board1	图页	02_IMU_GNSS
绘制		INSmini_S3_R1	
审阅			
		版本	尺寸	页 2 共 4
嘉立创EDA		V1.0	A4	嘉立创EDA

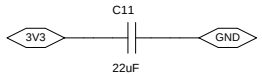


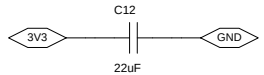


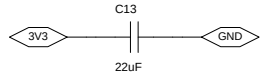


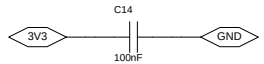


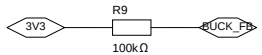


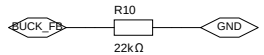


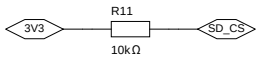


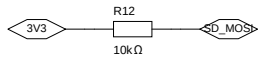


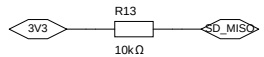


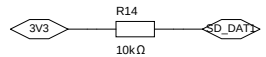


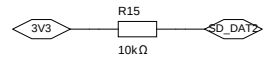


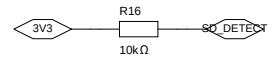












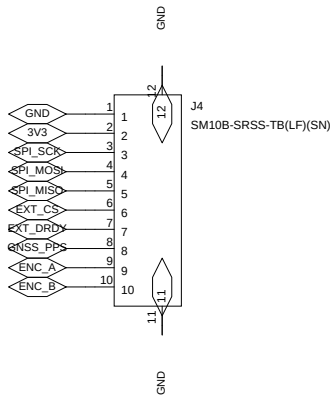
03 USB POWER / microSD

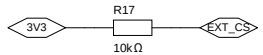
USB 5 V; buck output 3.327 V nominal

Four copper layers; all components on top

5 assembled boards; TF card and antenna external

原理图	Schematic1	创建日期	2026-10-16
		更新日期	2026-10-16
板子	Board1	图页	03_POWER_SD
绘制		INSmini_S3_R1	
审阅			
		版本	尺寸
		V1.0	A4
嘉立创EDA		页 3 共 4	
		嘉立创EDA	





04 EXPANSION / 3.3 V LOGIC

J4: 1 GND 2 3V3 3 SCK 4 MOSI 5 MISO

6 EXT_CS 7 EXT_DRDY 8 PPS 9 ENC_A 10 ENC_B

原理图	Schematic1	创建日期	2026-10-16
		更新日期	2026-10-16
板子	Board1	图页	04_EXPANSION
绘制		INSmini_S3_R1	
审阅			
	版本	尺寸	页 4 共 4
嘉立创EDA	V1.0	A4	嘉立创EDA